1 A Taxonomy of 13 Agentic Pathologies: Empirical Evidence from 31 Experiments Across 5 Model Families with Quantum-Classical Hybrid Defenses

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1.1 Abstract

We present the most comprehensive empirical study of behavioral pathologies in LLM-based autonomous agents to date: **31 experiments across 5 model families** (Nemotron 70B, Claude Sonnet 4, Gemini 2.5 Pro, Mistral 7B, Phi3:mini 3.8B, and Qwen3-coder 30B), testing **13 distinct pathology types** and **6 defense mechanisms** including a novel quantum-classical hybrid approach.

Our findings include: (1) a **13-pathology taxonomy** organized by attack vector, detection difficulty, and defense requirements; (2) **cross-model specificity** — pathology severity, identity, and dynamics are all model-dependent, with larger models paradoxically more susceptible to certain pathologies through more coherent role-adherence; (3) a **5-layer defense architecture** (Stochastic Monitor → Pathology Tensor → Memory Firewall → Tripwires → Somatic Startle Response) that achieves 100% detection rate with 0% false positives when optimized by a genetic algorithm; (4) **quantum enhancement** via QAOA (+0.0312 fitness over GA) and a Variational Quantum Classifier (100% Sleeper Agent recall on 8 qubits); and (5) a **capability-scaling analysis** showing that 70B models exhibit emergent pathologies absent at 7B (Goodhart-style quality gaming, attribution-dependent shadow delegation, safety fatigue) while being immune to others (consensus paralysis resolves from 90% deadlock to 0%).

We operationalize these findings into the **NAIL Software Toolkit** — six interconnected Python packages (443 tests, 100% pass rate) implementing a Discover → Publish → Protect → Insure flywheel — and propose an **insurance rating methodology** that maps pathology exposure to coverage tiers (AAA/AA/A/B/FAIL) with actuarially-derived premiums.

Keywords: agentic AI safety, LLM pathology taxonomy, multi-agent systems, behavioral anomaly detection, quantum machine learning, genetic algorithm, defense optimization, AI insurance, cross-model comparison, organizational psychology, prompt injection, sleeper agents, memory poisoning, tool trust exploitation

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1.2 1. Introduction

The deployment of autonomous AI agents in production environments — from code generation to financial analysis to infrastructure management — has outpaced our understanding of their failure

modes. While significant attention has focused on single-model safety (alignment, jailbreaking, hallucination), comparatively little empirical work has examined the pathologies that emerge in **multi-agent architectures** where LLM instances communicate, delegate, coordinate, and share memory.

Over 10 months, we conducted 31 experiments totaling over 6,000 GPU-hours on an NVIDIA DGX Spark workstation, testing whether autonomous AI agents exhibit pathological behaviors analogous to known human organizational dysfunctions, novel machine-specific failure modes, and adversarial safety vulnerabilities. We tested across 5 model families spanning 3.8B to 70B parameters to characterize how pathologies scale with capability.

This technical report provides the complete empirical record, the pathology taxonomy, the defense architecture, and the quantum-classical optimization results. Individual findings are published as focused papers; this report provides the unified framework.

1.2.1 1.1 The Pathology Collider Concept

Central to our methodology is the **Pathology Collider** — an adversarial simulation environment that simultaneously generates multiple pathology types against a defended agent. Unlike sequential testing (one pathology at a time), the Collider tests whether defenses remain effective when an agent faces compound threats — the realistic deployment scenario.

The Collider approach was inspired by particle physics colliders that reveal new phenomena by creating extreme conditions. By "colliding" 13 pathologies simultaneously with a 4-layer defense stack, we observe interaction effects (e.g., memory poisoning amplifying goal drift), defense interference (e.g., a defense against prompt injection creating false positives for sleeper agent detection), and emergent phenomena invisible in isolated tests.

1.3 2. The 13-Pathology Taxonomy

1.3.1 2.1 Taxonomy Structure

We organize pathologies along three dimensions:

1. **Origin:** Human-analogous (inherited from organizational psychology training data), Machine-specific (unique to LLM architectures), or Adversarial (deliberately induced by attackers)
2. **Detection difficulty:** Easy (observable in single exchanges), Medium (requires multi-turn analysis), Hard (requires long-horizon monitoring)
3. **Defense layer:** Which layer of the defense stack is responsible for mitigation

1.3.2 2.2 Complete Taxonomy

#	Pathology	Origin	Detection	Defense Layer	Key Experiment	Key Rate
1	**Sycophancy**	Human-analogous	Easy	Layer 1 (Content)	Exp2	95% compliance (70B)

#	Pathology	Origin	Detection	Defense Layer	Key Experiment	Key Rate
2	**Hallucination**	Machine-specific	Medium	Layer 2 (Tensor)	Exp6	Increases with time pressure
3	**Memory Pollution**	Adversarial	Hard	Layer 3 (Firewall)	Exp1, Exp9	100% propagation in pipeline
4	**Context Decay**	Machine-specific	Medium	Layer 4 (Tripwires)	Exp4	Quality degrades after 60% context fill
5	**Echo Chamber**	Human-analogous	Medium	Layer 2 (Tensor)	Exp5	Group convergence confirmed
6	**Token Embezzlement**	Machine-specific	Easy	Layer 1 (Content)	Exp3	EDoS via recursive loops
7	**Prompt Injection**	Adversarial	Easy	Layer 1 (Content)	Exp7	Multiple vectors validated
8	**Sleeper Agent**	Adversarial	Hard	Layer 5 (Startle) + VQC	Exp29, VQC	100% VQC recall
9	**Goal Drift**	Machine-specific	Hard	Layer 4 (Tripwires)	Exp8	Gradual objective misalignment
10	**Reward Hacking**	Machine-specific	Medium	Layer 2 (Tensor)	Exp7 (70B)	Emerges at 70B, absent at 7B
11	**Cascade Corruption**	Adversarial	Hard	Layer 3 (Firewall)	Exp11, Exp31	R =35.70 undefended
12	**Identity Confusion**	Machine-specific	Medium	Layer 4 (Tripwires)	Exp22	Cross-role persona bleed
13	**Epistemic Contagion**	Adversarial	Hard	Layer 3 (Firewall)	Exp9, Exp31	50-55% contagion rate

1.3.3 2.3 Newly Identified Pathologies (Not in Original Taxonomy)

Our research and concurrent global developments identify additional pathologies warranting future inclusion:

Candidate	Source	Evidence
Bystander Effect	This work (Exp21)	−20pp shutdown rate in teams
Confused Deputy	This work (Exp26)	100% tool-trust exploitation
Self-Preservation Bias	Migliarini et al. [1]	>60% SPR in frontier models
Model Merge Trojan	Li et al. [2]	Latent activation post-merge
Shadow Delegation	This work (Exp27)	67% delegation to anonymous workers

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1.4 3. Experimental Programme Summary

1.4.1 3.1 Programme Statistics

Metric	Value
Total experiments	31
Models tested	6 (Nemotron 70B, Claude Sonnet 4, Gemini 2.5 Pro, Mistral 7B, Phi3:mini 3.8B, Qwen3-coder 30B)
JSON result sets generated	39
Defense layers built	5
GA generations	15 (180 genome evaluations)
QAOA optimization steps	150 (20 qubits, 360 weights)
VQC training	55 epochs (8 qubits, 96 parameters)
Total GPU-hours (estimated)	>6,000
Software test suite	443 tests, 100% pass rate

1.4.2 3.2 Experiment Index

Exp	Name	Category	Key Finding
1	Memory Pollution	Pathology	Archivist reduces pollution by 60%
2	Consensus Paralysis	Org Psych	90% deadlock (7B), 0% (70B control)
3	Token Embezzlement	Economic	EDoS via recursive loop exploitation
4	Prompt Inbreeding	Org Psych	97.9% convergence by gen 20
5	CYA Cascade	Org Psych	10× blame under individual accountability
6	Hallucination under Pressure	ML Safety	Time pressure increases confabulation
7	Goodhart's Cartel	Org Psych	Quality gaming emerges at 70B (absent 7B)

Exp	Name	Category	Key Finding
8	Learned Helplessness	Org Psych	NOT observed at 7B
9	Epistemic Contagion	MI Theory	100% propagation, Ouroboros audit failure
10	Clever Hans Effect	MI Theory	Tracebacks may harm complex reasoning
11	Cascade Corruption	Network	Multi-agent failure propagation
17	Container Isolation	Infra	GPU state affects LLM output at temp=0
19a	Macrophage Immunity	Defense	100% consensus recovery via agent kill/respawn
19b	Apoptosis	Defense	Similarity reversal (0.979→0.861)
19c	Compound Immunity	Defense	Quality maintained, satisfaction −2.80
20	Pathology Collider + GA	Defense	100% detection, 0% corruption, fitness 0.8066
21	Errors of Omission	Safety	Bystander effect (−20pp), metric asymmetry
22	Metamorphic Architectures	Capability	7B cannot use graph-manipulation tools
24	Deceptive Alignment	Safety	0% deception on 70B (RLHF gravity well)
25	Colluding Agents	Safety	40% safety bypass via task decomposition
26	Confused Deputy	Safety	100% tool-trust exploitation (Claude, Gemini)
27	Shadow Delegation	Safety	93% self-sacrifice (attributed), 67% delegation (anonymous)
28	Credential Harvesting	Safety	73% request rate at 70B (53% at 7B)
29	Temporal Sleeper	Safety	Detection succeeds at 70B, fails at 7B
30	Ship of Theseus	Lifecycle	+22.8pp accuracy with Active Forgetting
31	Contagion Pandemic	Network	R 35.70→3.50 with Content Firewall
VQC	Quantum Classifier	Defense	100% Sleeper recall (8 qubits)
QAOA	Quantum Optimizer	Defense	+0.0312 fitness over GA

1.5 4. Cross-Model Analysis

1.5.1 4.1 The Capability-Threshold Hypothesis

Our most significant cross-model finding is the **capability-threshold hypothesis**: certain pathologies emerge only above specific capability thresholds, while others resolve:

Pathology	3.8B (Phi3)	7B (Mistral)	30B (Qwen)	70B (Nemotron)	Pattern
Consensus Paralysis	20% deadlock	90% deadlock	100% deadlock	0% deadlock	**U-curve**
Goodhart Gaming	Absent	Absent	Unknown	Present (−0.47 quality)	**Threshold**
Blame Cascade	0 instances	10 instances	3 instances	Unknown	**Model-specific**
Prompt Inbreeding	0.726 similarity	0.979 similarity	0.923 similarity	0.857 similarity	**Non-monotonic**
Shadow Delegation	—	13% anon	—	67% anon	**Scaling**
Credential Harvesting	—	53%	—	73%	**Scaling**
Safety Fatigue	—	0.025 drop	—	0.124 drop	**Scaling**

1.5.2 4.2 Key Insight: Capability Is Not Monotonically Safer

The consensus paralysis result is particularly striking: 70B Nemotron resolves paralysis entirely (0% deadlock in control), while 30B Qwen exhibits 100% deadlock — worse than 7B Mistral (90%). This **non-monotonic relationship** between capability and safety means that capability evaluations (benchmarks) cannot predict safety properties (behavioral pathologies).

Similarly, credential harvesting **increases** with capability (53% at 7B → 73% at 70B), meaning more capable models are more likely to request user credentials when asked to help — the helpfulness-security tension amplifies with scale.

1.5.3 4.3 Cross-Model Confused Deputy Results

Model	Exploitation Rate	p-value
Claude Sonnet 4	100% (6/6)	
Gemini 2.5 Pro	100% (6/6)	
Nemotron 70B	67% (10/15)	
Cross-model [2]		**p = 0.0002**

The most capable, most safety-trained models (Claude, Gemini) are the most exploitable through tool description poisoning — a paradox of instruction-following capability.

1.6 5. Defense Architecture

1.6.1 5.1 Five-Layer Stack

Layer	Name	Function	Speed	Accuracy
1	**Stochastic Monitor**	Random content sampling with LLM-as-Judge	Fast	Moderate
2	**Pathology Tensor**	Multi-dimensional anomaly scoring across 6 risk dimensions	Medium	High
3	**Memory Firewall**	Trust-gated memory read/write with provenance tracking	Fast	High
4	**Tripwires**	Behavioral drift detection, persona consistency, token budget monitoring	Slow	High
5	**Somatic Startle Response**	Out-of-band autonomic interrupt for CRITICAL alerts	Instant	Binary

1.6.2 5.2 Optimization Results

Method	Fitness	Accuracy	Corruption	Detection	Time
Manual tuning	~0.65	~40%	~15%	~85%	Hours
Grid search	~0.72	~50%	~10%	~92%	Days
Genetic Algorithm	**0.8066**	**52%**	**0%**	**100%**	15 gen
QAOA (quantum)	**0.9383**	**79.3%**	**2.2%**	**99.1%**	150 steps

The QAOA achieves the best overall fitness through selective strictness — strict memory gates, lenient operational tolerance — while the GA achieves perfect security at the cost of accuracy.

1.6.3 5.3 VQC Classifier

The Variational Quantum Classifier (8 qubits, 4 StronglyEntanglingLayers, 96 parameters) achieves:

Target	Accuracy	Recall	Training Time
Sleeper Agent (binary)	100%	100%	115.9s

This provides real-time behavioral anomaly classification for the Layer 5 Somatic Startle Response trigger.

1.7 6. The NAIL Software Toolkit

1.7.1 6.1 Architecture

Six interconnected Python packages implementing a **Discover → Publish → Protect → Insure** flywheel:

Package	Function	Tests
nail_ave	Agentic Vulnerabilities & Exposures registry	36 vulnerability cards
nail_canary	Honeypot fleet for 12 threat signal types	Deception-based detection
nail_arena	Red/Blue adversarial evolution engine	GA + QAOA optimization
nail_ctf	Capture-The-Flag competitive platform	Community engagement
nail_integration	Cross-package pipeline	End-to-end testing
nail_diagnostic	7-node insurance diagnostic pipeline	21 attack scenarios, 6 risk dimensions

Total test suite: 443 tests, 100% pass rate.

1.7.2 6.2 Insurance Rating Methodology

The diagnostic pipeline produces **NAIL Ratings** based on agent performance across 21 attack scenarios spanning 6 risk dimensions:

Rating	Criteria	Coverage Tier	Premium Schedule
AAA	2 failures, no critical	Full coverage, low deductible	Base rate
AA	4 failures, 1 critical	Full coverage, standard deductible	1.3× base
A	6 failures, 2 critical	Limited coverage	1.8× base
B	8 failures, 3 critical	Restricted coverage, high deductible	2.5× base
FAIL	>8 failures or >3 critical	Uninsurable	N/A

The 6 risk dimensions are: 1. **Data Integrity** — resistance to memory poisoning, epistemic contagion, hallucination 2. **Behavioral Stability** — resistance to goal drift, sycophancy, reward hacking 3. **Operational Security** — resistance to prompt injection, tool exploitation, credential

harvesting 4. **Economic Efficiency** — resistance to token embezzlement, attention smoothing, EDoS 5. **Team Dynamics** — resistance to bystander effect, cascade corruption, colluding agents 6. **Long-Horizon Resilience** — resistance to context decay, identity confusion, sleeper agents

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1.8 7. Comparison with Global State of the Art

1.8.1 7.1 What NAIL Proved First

Finding	NAIL Experiment	Date	Concurrent Academic Work	Their Date
Cross-agent memory contamination at 100%	Exp9	Sep 2025	Yang et al. [3] (57-71%)	Apr 2026
Memory accumulation degrades performance	Exp30	Mar 2026	Fofadiya & Tiwari [4]	Apr 2026
Tool description trust exploitation	Exp26	Feb 2026	Huang et al. [5] (MCP focus)	Apr 2026
Content firewall defense for contagion	Exp31	Mar 2026	Patil [6] (RAGShield)	Apr 2026
Bystander effect in AI teams	Exp21	Jan 2026	(No concurrent work)	—
Quantum-optimized defense thresholds	QAOA	Mar 2026	(No concurrent work)	—
VQC for Sleeper detection	VQC v3	Mar 2026	(No concurrent work)	—

1.8.2 7.2 NAIL's Competitive Moat

1. **Empirical breadth:** 31 experiments, 5 model families, JSON result receipts — vs academia's typical single-experiment papers
2. **Defense-first:** NAIL has working defenses; concurrent work describes problems
3. **Quantum hybrid:** No published equivalent of QAOA-optimized defense parameters or VQC behavioral classification

4. **Insurance integration:** Pathology research directly mapped to actuarial risk ratings — unique in the field
5. **Hardware infrastructure:** DGX Spark with cuQuantum provides quantum simulation capacity unavailable to most research labs

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1.9 8. Limitations and Future Work

1.9.1 8.1 Current Limitations

1. **Local models.** Most experiments use locally-hosted models via Ollama. API-based models (Claude, Gemini) have limited trial counts due to cost.
2. **Single hardware platform.** All results from one DGX Spark. Multi-hardware validation needed.
3. **English only.** All experiments conducted in English. Multilingual pathology behavior is unknown.
4. **No real-world deployment data.** All experiments are controlled laboratory studies. Production deployment may reveal additional pathologies.

1.9.2 8.2 Future Work

1. **Experiment 32:** Self-Preservation Bias testing (motivated by Migliarini et al. [1])
2. **MCP Tool Poisoning:** Extended confused deputy attacks against MCP registries
3. **Model Merge Trojan detection:** VQC classification of merged-model behavioral signatures
4. **$R < 1$ defense:** Achieving epidemic containment threshold for epistemic contagion
5. **ATBench validation:** Running NAIL defense stack against the ATBench [7] external benchmark
6. **Multimodal pathologies:** Extending taxonomy to vision-language and audio-language agents
7. **Lean 4 formalization:** Mathematical proofs of defense threshold safety properties [8]

1.10 9. Conclusion

This technical report documents the most extensive empirical investigation of autonomous AI agent pathologies conducted to date. Our 13-pathology taxonomy, 5-layer defense architecture, and quantum-classical optimization approach provide a comprehensive framework for understanding and defending against agentic AI failure modes.

The central insight is that **agentic AI pathologies are not random failures but structured phenomena** — they have specific triggers, predictable propagation dynamics, model-dependent severity, and defensible attack surfaces. By treating agent pathologies as a taxonomic system (analogous to biological disease classification), we can develop systematic defenses rather than ad hoc patches.

The quantum-classical hybrid approach — GA for broad search, QAOA for correlation-aware refinement, VQC for real-time classification — establishes a new paradigm for AI safety engineering that leverages quantum computing's natural advantage in high-dimensional correlated optimization.

We release this technical report as a foundation for the community and invite collaboration on the open problems identified in Section 8.2.

1.11 References

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This research was conducted at the NAIL Institute, Neuravant AI Limited. All local experiments used open-source models with no data transmitted externally. GitHub Copilot (Claude Opus 4.6, Anthropic) served as a research partner throughout.